

# Building LLM Reasoners Project Check-in

Gaze-Guided Acceleration for On-Device SmolVLM Inference

Julia Gontijo Lopes

Sara Vargas Martínez

## 1 Introduction

Vision-language models have shown strong performance on multimodal tasks, but they remain too slow for real-time use in many settings. The main bottleneck is the conversion of images into long sequences of visual tokens, which makes autoregressive decoding expensive in both memory and latency—a problem that persists even in smaller models like SmolVLM. In this project, we ask whether eye gaze, already available from wearable devices, can be used to make inference more efficient. Since gaze naturally reflects what a user is attending to, it can serve as a proxy for visual relevance. We propose a gaze-guided pruning method that removes visual tokens from regions far from the gaze point before decoding, reducing sequence length and inference cost without modifying the model architecture or requiring retraining. Initial results show roughly  $1.4\text{--}1.7\times$  speedup while preserving high similarity to the full model’s outputs.

## 2 Related Work

Reducing inference cost in vision-language models is an active area of research, as computational cost grows quickly with the number of visual tokens.

Our work builds on SmolVLM [1], a compact model family designed for efficient multimodal inference that can fit on edge hardware after quantization. Despite its small size, SmolVLM still relies on autoregressive decoding, which remains a bottleneck in interactive settings. This makes it a natural target for inference-time acceleration methods that do not require retraining.

Token pruning is one common approach to reducing this cost, where less informative visual tokens are removed during inference. SparseVLM [3] prunes visual tokens based on attention signals without retraining, showing that substantial compute can be removed while largely preserving task performance.

Eye gaze offers a different kind of signal. GazeVLM [4] shows that gaze can guide a model toward the most task-relevant visual content, improving grounding and prediction quality. However, it treats gaze as a semantic cue rather than a computational one. Our work combines these motivations: we use gaze not to improve understanding accuracy, but to make on-device generation faster by determining which visual regions to keep before decoding. We ground this setup in EgoExo4D [2], which provides the egocentric video and synchronized gaze annotations needed to make this approach practical.

## 3 Data Collection and Preprocessing

A significant portion of our current effort went into building a reliable data pipeline for egocentric video and gaze synchronization. We use the EgoExo4D dataset [2], which provides paired egocentric video and gaze annotations from wearable devices. After resolving licensing constraints and obtaining access, we developed a preprocessing pipeline to construct aligned input samples. This involved extracting frames at a fixed

sampling rate (1 frame per second), aligning gaze timestamps with corresponding frames, and normalizing gaze coordinates into image space.

For preliminary evaluation, we use two EgoExo4D sequences: `iiith_cooking_108_5` (64 frames) and `nus_cpr_27_3` (50 frames), selected to cover different activity types and test generalization across environments.

## 4 Methodology

### 4.1 Problem Setup

Our goal is to reduce the number of visual tokens processed by SmolVLM during inference while preserving output quality. In order to do this we developed a gaze guided pruning module that operates after visual encoding but before the language model decoder. This is done using the tiling structure SmolVLM uses, since every frame inputted is represented as a set of fixed visual tiles, one global tile capturing coarse scene context, and multiple local tiles encoding higher-resolution regions. Since each tile produces a fixed number of tokens, we apply pruning at the tile level rather than on individual tokens.

### 4.2 Gaze-Conditioned Tile Scoring and Pruning

For each frame, we use a synchronized gaze point from the EgoExo4D dataset recordings, normalized into image coordinates. To discern the importance of the tile we calculate a relevance score for each tile based on its spatial proximity to the gaze point:

$$s_i = \exp\left(-\frac{\|\mathbf{c}_i - \mathbf{g}\|_2^2}{2\sigma^2}\right),$$

where  $\mathbf{c}_i$  is the tile center and  $\mathbf{g}$  is the gaze location. Tiles are ranked by this score, and only the top- $k$  fraction (controlled by the keep ratio) is retained. The global tile is always kept to preserve coarse scene context and all visual tokens from pruned tiles are removed, producing a shorter visual sequence for the decoder which is our goal.

Figures 1 and 2 show examples of the resulting pruning pattern, which is spatially structured around the fixation point.

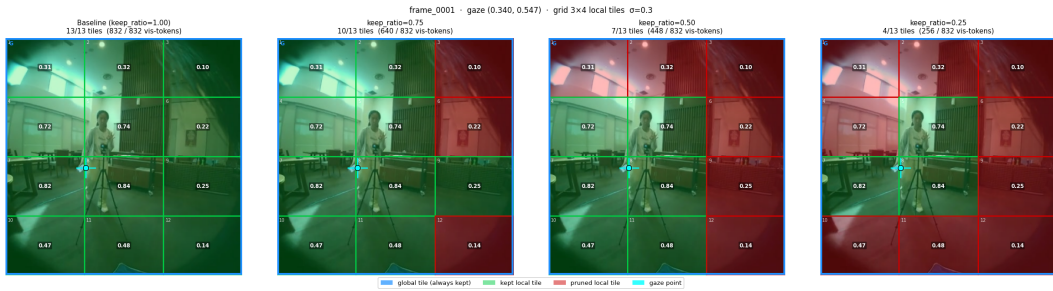


Figure 1: Gaze-guided pruning on `nus_cpr_27_3`. Green tiles are retained, red tiles are removed, and the blue point indicates gaze fixation.

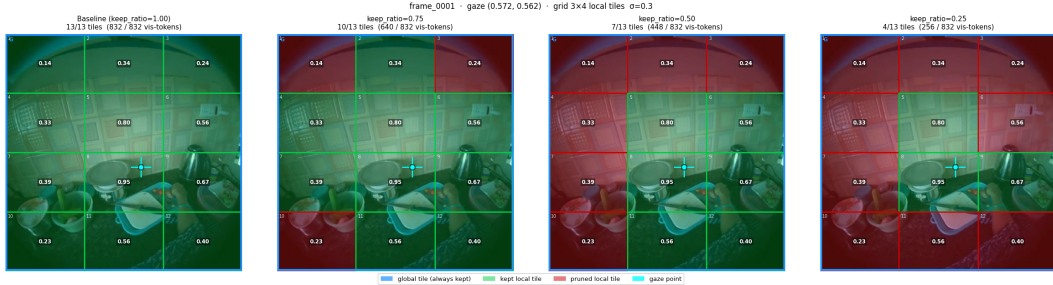


Figure 2: Gaze-guided pruning on `iiith_cooking_108_5`. The method preserves regions near gaze while removing peripheral tiles.

## 5 Preliminary Results

To evaluate this pruning module we evaluate on two sequences of video frames from EgoExo4D with real gaze annotations (114 frames total), using the prompt “*What is the person doing?*”. The results we reported both look into the inference efficiency to see if there is an observable speedup from the baseline and the semantic similarity to evaluate if there was any loss of inference power in the output after pruning (this is compared to the non-pruned model output as a baseline).

### 5.1 Efficiency

Tables 1 and 2 summarize how each keep ratio maps to tile and token counts and Tables 3 and 4 report their runtimes.

Table 1: Pruning configuration for `iiith_cooking_108_5`.

| keep_ratio      | Tiles kept | Visual tokens |
|-----------------|------------|---------------|
| 1.00 (baseline) | 17 / 17    | 1088          |
| 0.75            | 13 / 17    | 832           |
| 0.50            | 9 / 17     | 576           |
| 0.25            | 5 / 17     | 320           |

Table 2: Pruning configuration for `nus_cpr_27_3`.

| keep_ratio      | Tiles kept | Visual tokens |
|-----------------|------------|---------------|
| 1.00 (baseline) | 17 / 17    | 1088          |
| 0.75            | 13 / 17    | 832           |
| 0.50            | 9 / 17     | 576           |
| 0.25            | 5 / 17     | 320           |

From these results we can see that reducing visual tokens consistently reduces runtime, even if the improvements aren’t linear with pruning. At the most aggressive setting of 1088 to 320 tokens, we see the speedups ranged around 1.4–1.7 $\times$ . This suggests inference cost is not dominated by visual token processing alone and that there is likely some overheads such as model invocation, decoding initialization that contribute to total latency.

Table 3: Inference efficiency for `iiith_cooking_108_5`.

| <b>keep_ratio</b> | <b>Time (s)</b>   | <b>Speedup</b>  |
|-------------------|-------------------|-----------------|
| 1.00 (baseline)   | $0.466 \pm 0.103$ | 1.00            |
| 0.75              | $0.302 \pm 0.037$ | $1.57 \pm 0.50$ |
| 0.50              | $0.307 \pm 0.041$ | $1.54 \pm 0.33$ |
| 0.25              | $0.306 \pm 0.046$ | $1.55 \pm 0.37$ |

Table 4: Inference efficiency for `nus_cpr_27_3`.

| <b>keep_ratio</b> | <b>Time (s)</b>   | <b>Speedup</b>  |
|-------------------|-------------------|-----------------|
| 1.00 (baseline)   | $0.463 \pm 0.058$ | 1.00            |
| 0.75              | $0.328 \pm 0.035$ | $1.42 \pm 0.21$ |
| 0.50              | $0.288 \pm 0.048$ | $1.64 \pm 0.32$ |
| 0.25              | $0.281 \pm 0.046$ | $1.69 \pm 0.36$ |

## 5.2 Semantic Preservation

The semantic quality was measured using BERTScore F1 so we could capture if there was any degradation in the outputs after omitting certain input. The score is comparing our pruned output against the full-model baseline. In Table 5 we can see the performance of both models and we observe that similarity remains high across all pruning levels and degrades very gradually which suggests that gaze-guided pruning removes mostly redundant visual information while retaining task-relevant regions.

Table 5: Summary of BERTScore F1 across datasets.

| <b>keep_ratio</b> | <b>iiith_cooking</b> | <b>nus_cpr</b> |
|-------------------|----------------------|----------------|
| 1.00 (baseline)   | 1.000                | 1.000          |
| 0.75              | 0.967                | 0.959          |
| 0.50              | 0.952                | 0.941          |
| 0.25              | 0.947                | 0.938          |

Overall, moderate pruning (`keep_ratio` = 0.75) offers the best efficiency-quality tradeoff. Further reduction yields diminishing speed gains while semantic quality holds up well.

## 6 Next Steps

A key limitation of the current approach is that pruning is that it is task-agnostic, the same spatial strategy is applied regardless of whether the question calls for local object recognition, action recognition, or global scene understanding. Our next steps aim to address this.

We plan to group questions by reasoning type at the object-level, action-level, and scene-level to better characterize where gaze-guided pruning is robust and where it breaks down. This will also let us test whether different keep ratios are optimal for different task categories, since some may tolerate more aggressive pruning than others. We also want to incorporate the text query directly into the pruning decision, so that gaze and language jointly determine which regions to retain.

Once the pruning strategy is more settled, we plan to explore fine-tuning SmolVLM with gaze as an explicit input signal, rather than using it only as a post-hoc filter on visual tokens. The idea is to let the model learn to integrate gaze during training, which may produce better grounding and more robust token selection than our current inference-only approach. For evaluation, we plan to include GazeVLM [4] as a gaze-aware

baseline, which allows for a fairer comparison than the unpruned model: both systems receive gaze as input, but use it for different purposes. Finally, we aim to validate the full pipeline in our Unity deployment on real wearable hardware, which is the target setting motivating this work.

## References

- [1] Andrés Marafioti, Orr Zohar, Miquel Farré, Merve Noyan, Elie Bakouch, Pedro Cuenca, Cyril Zakka, Loubna Ben Allal, Anton Lozhkov, Nouamane Tazi, Vaibhav Srivastav, Joshua Lochner, Hugo Larcher, Mathieu Morlon, Lewis Tunstall, Leandro von Werra, and Thomas Wolf. *SmolVLM: Redefining Small and Efficient Multimodal Models*. arXiv preprint arXiv:2504.05299, 2025.
- [2] Kristen Grauman et al. *Ego-Exo4D: Understanding Skilled Human Activity from First- and Third-Person Perspectives*. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [3] Yuan Zhang, Chun-Kai Fan, Junpeng Ma, Wenzhao Zheng, Tao Huang, Kuan Cheng, Denis Gudovskiy, Tomoyuki Okuno, Yohei Nakata, Kurt Keutzer, and Shanghang Zhang. *SparseVLM: Visual Token Sparsification for Efficient Vision-Language Model Inference*. arXiv preprint arXiv:2410.04417, 2024.
- [4] Athul M. Mathew, Haithem Hermassi, Thariq Khalid, Arshad Ali Khan, and Riad Souissi. *GazeVLM: A Vision-Language Model for Multi-Task Gaze Understanding*. arXiv preprint, 2025.